Chapter 6 Sec 78 , 80-83 More about Residues and Poles

The three types of isolated singular points.

Just a reminder

An *isolated singular point* is a point at which $f(z)$ is not analytic and there is a deleted neighborhood around that point where $f(z)$ is analytic.

The types..



1. Removable singular points: We know that around an isolated singular point we have a Laurent Series for $f(x)$ given by

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \qquad R_1 < |z - z_0| < R_2$$

Where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \qquad n = 0,1,2, \ldots \qquad b_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{-n+1}} dz \quad n = 1,2, \ldots.$$

If every b_n is zero so that

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

We call z_0 a a removable singularity. What we really mean by this is that we were able to assingn a value of a_0 to $f(z)$ at the point of singularity to make it analytic. Let's look at an example.

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Example of a removable singularity

$$f(z) = \frac{1 - \cosh(z)}{z^2}$$

To help us along recall that the Taylor series for $\cosh(z)$ is given by: $\cosh(z) = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \cdots \quad |z| < \infty$

So since the $b_n's$ all dissappear in the series we can redefine $f(z)$ as

$$f(z) = \begin{cases} \frac{1 - \cosh(z)}{z^2} & if\ z \neq 0 \\ -\frac{1}{2} & if\ z = 0 \end{cases}$$

To make it analytic, thus removing the singularity!

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

2. Essential Singularity: If $f(z)$ is not analytic at z_0 and we have an infinite number of NON-ZERO $b_n's$ in the laurant series for $f(z)$ we call z_0 an *essential singularity*

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n} \qquad R_1 < |z-z_0| < R_2$$

Where

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{n+1}}dz \qquad n = 0,1,2,\ldots \qquad b_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{-n+1}}dz \quad n = 1,2,\ldots.$$

So the singularity is essential if we have an infinite number of $b_n's$ in our Laurent series, Let's look at an example:

$$f(z) = e^{\frac{1}{z}}$$

This has an essential singularity at $z = 0$ let's see why:

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

3. Pole of order m: If at some number m all of the b_n's =0 for $n > m$ in the laurant series for $f(z)$

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \qquad R_1 < |z - z_0| < R_2$$

Where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \qquad n = 0,1,2, \ldots \qquad b_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{-n+1}} dz \quad n = 1,2, \ldots.$$

That is $b_m \neq 0$ and $b_{m+1} = b_{m+2} = \cdots = 0$ Then we say that $f(z)$ has a *pole of order m* at z_0

That means out expansion might look like

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} =$$

Example: $f(z) = \frac{1}{z^2(1 - z)}$ $0 < z < 1$ Has a pole of order 2 at $z = 0$

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Residues at Poles

Theorem: Let z_0 be an isolated singular point of a function $f(z)$ the following two statements are equivlent.

1. z_0 is a pole of order m
2. $f(z)$ can be written in the form

$$\frac{\phi(z)}{(z-z_0)^m} \qquad m = 1,2 \ldots$$

Where $\phi(z)$ is analytic and nonzero at z_0

More over if statements 1,2 are true

$$Res_{z=z_0} f(z) = \phi(z_0) \text{ When } m = 1$$

$$Res_{z=z_0} f(z) = \frac{\phi^{(m-1)}(z_0)}{(m-1)!} \qquad \text{When } m = 2,3,\cdots$$

Proof assume 1), Then we know $f(z)$ has a laurant series expansion with the $b_n's$ stopping at some b_m.

Choose $\phi(z) = \begin{cases} (z-z_0)^m f(z) & when \quad z \neq z_0 \\ b_m & when \quad z = z_0 \end{cases}$ If we write out the series for $\phi(z)$ we get..

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

So

$$\phi(z) = b_m + b_{m-1}(z - z_0) + \cdots b_1(z - z_0)^{m-1} + \sum_{n=0}^{\infty}(z - z_0)^{m+n}$$

Which is indeed analytic on the entire disk $|z - z_0| < R$

If you divide this by $(z - z_0)^m$ we end up with the Laurent series for $f(z)$

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Assume 2 $f(z)$ can be written in the form

$$f(z) = \frac{\phi(z)}{(z - z_0)^m}$$

Where $\phi(z)$ is analytic and nonzero at z_0

Since we know that $\phi(z)$ is analytic it has a taylor series given by

$$\phi(z) = \sum_{n=0}^{\infty} \frac{\phi^{(n)}(z_0)}{n!}(z - z_0)^n$$

When $|z - z_0| < R_0$

Let's write out the first m terms of the sum and then divide by $(z - z_0)^m$ to look at the series for $f(z)$

So

$$f(z) = \frac{\phi(z_0)}{(z - z_0)^m} + \frac{\phi'(z_0)}{1!\,(z - z_0)^{m-1}} + \cdots \frac{\phi^{(m-1)}(z_0)}{(m-1)!\,(z - z_0)} + \sum_{n=m}^{\infty} \frac{\phi^n(z_0)}{n!}(z - z_0)^{n-m}$$

So we see that $f(z)$ has a pole of order m at z_0!

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Examples:

The function $f(z) = \frac{z+4}{z^2+1}$ Can be written as $f(z) = \frac{\phi(z)}{z-i}$ Where $\phi(z)$ is analytic at $z = i$

Find $\phi(z)$

Use our theorem to find the residue of $f(z)$ below at $z = i$

$$Res_{z=z_0} f(z) = \frac{\phi^{(m-1)}(z_0)}{(m-1)!}$$

The function $f(z) = \frac{z^3+2z}{(z-i)^3}$

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Zeros of Order m

We know that if a function f is analytic at a point z_0 then all of it's derivatives exist, Suppose that

$$f(z_0) = f'(z_0) = f''(z_0) = \cdots f^{(m-1)}(z_0) = 0 \text{ and } f^{(m)}(z_0) \neq 0$$

We then call z_0 a zero of order m

Theorem: Let f be analytic at z_0 then the following statements are equivlent.

1. $f(z)$ has a zero of order m at z_0.
2. there is a function g , wich is analytic and nozero at z_0 such that

$$f(z) = (z - z_0)^m g(z)$$

Proof: 1 implies 2 To show that 1 implies 2 we can just write out the taylor series for f and realize that all terms a_n for $n < m$ are zero by the assumption of 1 and then just factor out $(z - z_0)^m$

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Theorem: Let f be analytic at z_0 then the following statements are equivlent.

1. $f(z)$ has a zero of order m at z_0.
2. there is a function g , wich is analytic and nozero at z_0 such that $f(z) = (z - z_0)^m g(z)$

2 implies 1

To prove 2 implies 1 we just need to write out the taylor series for $g(z)$ multiply by $(z - z_0)^m$ and use the fact that taylor series are unique.

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Theorem: Given a function f and a point z_0, suppose that

a) $f\ is\ analytic\ at\ z_0$
b) $f(z_0) = 0\ but\ f(z) is\ not\ identically\ equal\ to\ zero\ in\ any\ neighbor\ hood\ of\ z_0$

Then $f(z) \neq 0\ \ throughout\ some\ delete\ neighborhood\ 0 < |z - z_0| < \epsilon\ of\ z_0$



Proof: $f(z_0) = 0$ but the fact that $f(z)$ is not identically zero means that not all of it's derivatives can be zero in some neighborhood of z_0 if so then the taylor series which is equal to f would be identically zero which would mean f was

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \frac{f^{(m)}(z_0)}{m!}(z - z_0)^m + \cdots$$

So let m be the first non zero derivative then z_0 is a zero of order m this means that according to the theorem we just proved

$$f(z) = (z - z_0)^m g(z)$$

For some $g(z)$ where $g(z) \neq 0$ is some ϵ neighborhood of z_0, a theroem we had from long ago Consequently this means that if we are not at z_0 then $f(z) \neq 0$ in the neighborhood of $0 < |z - z_0| < \epsilon$

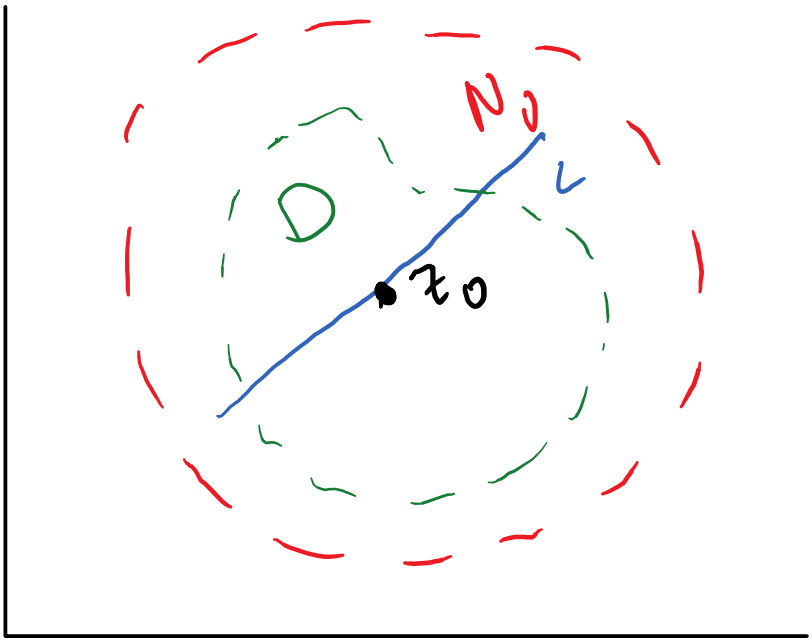
Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Another Theorem :)

Theorem: Given a function f and a point z_0 suppose that

1. f *is analytic throughout a neighborhood* N_0 *of* z_0
2. $f(z) = 0$ *at each point* z *of a domain* D *or line segment* L *containint* z_0

Then $f(z) = 0$ in N_0 *that is* , $f(z)$ *is identically equal to zero throughought* N_0



Proof: Assume not, Then according to the previous theorem there is a deleted neighborhood around z_0 where $f(z) \neq 0,$ Of course this contradicts the the assumption that $f(z)$ is zero throughought D This means it indeed need to be identically zero in N_0

Chapter 6 Sec 78 , 80-83 More about Residues and Poles

Zeros and Poles

There are some connections between zeros and poles of functions which we will explore below.

Theorem: Suppose that

$$a)\ two\ function\ p\ and\ q\ are\ analtic\ at\ a\ point\ z_0$$
$$b)\ p(z_0) \neq 0\ and\ q\ has\ a\ zero\ of\ order\ m\ at\ z_0$$

$$then\ the\ quuotient \frac{p(z)}{q(z)}\ has\ a\ pole\ of\ order\ m\ at\ z_0$$

Theorem: $Let\ two\ function\ p\ and\ q\ be\ analytic\ at\ a\ point\ z_0. If \quad p(z_0) \neq 0, q(z_0) = 0\ \ and\ q'(z_0) \neq 0$

Then z_0 is a simple pole of $\frac{p(z)}{q(z)}\ \ and$

$$Res_{z=z_0}\ \frac{p(z)}{q(z)} = \frac{p(z_0)}{q'(z_0)}$$

Homework Example : Show that $z = 0$ is a simple pole of $f(z) = \csc(z) = \frac{1}{\sin(z)}$ and that the residue at $z_0 = 0$ is 1.